

Multiple Choice (10 marks)

1. If a, b, and c are integers, which of the following conditions is sufficient to guarantee that the expression $a < c \parallel a < b \ \&\& \ ! (a == c)$ evaluates to true?
 - (a) $a < c$
 - (b) $a < b$
 - (c) $a > b$
 - (d) $a == b$
2. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
3. What is the value of this python expression: $4 + 3 \% 2$?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
4. In Python 3, let $r = \text{lambda } q: q * 2$. What is $r(3)$?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
5. Which of the following spreadsheet functions would be most useful for detecting improbably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row

True / False (10 marks)

Answer True or False

6. True or False: A group of cookies stored by the user's web browser has the greatest potential for security.
7. True or False: In Python 3, the expression 'abc'[-1] returns the last character, which is 'c'.
8. True or False: In Python3, the minimum value of the list l = [1,2,3,4] is 1.
9. True or False: In Python 3, the isupper() function checks if all characters in a string are uppercase.
10. True or False: Python supports automatic garbage collection to manage memory efficiently.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to convert polar coordinates to rectangular coordinates.
12. Write a function to check whether the given number is undulating or not.

End of Paper